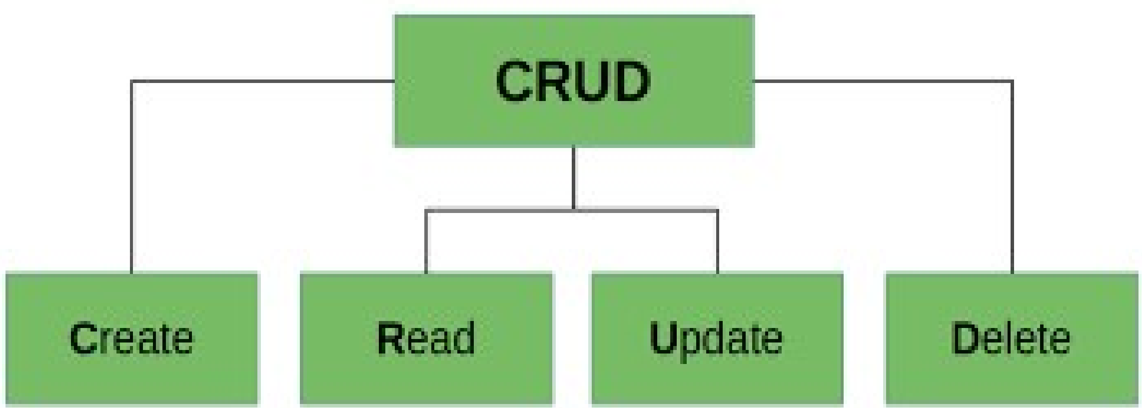
JDBC CRUD Operation

What is CRUD :-

- JDBC allows Java programs to **interact with databases** using **SQL commands.**
- JDBC Java Database Connectivity **enables Java applications to perform CRUD .**
- In JDBC there are CRUD (Create, Read, Update, Delete) operations on **relational databases.**
- These operations correspond to the **fundamental ways data is managed** within a database.



CRUD =

- **C** → Create (INSERT)
- **R** → Read (SELECT)
- **U** → Update (UPDATE)
- **D** → Delete (DELETE)

Create :-

- This operation is used to **add new records (rows)** into a database table.
- In JDBC, we use an **INSERT SQL statement** to perform this action.
- The INSERT query is executed using either a **Statement** or a **PreparedStatement** object.
- **PreparedStatement** is better because:
    - It protects from **SQL injection** (more secure).
    - It gives **better performance** when running the same query many times with different data.

- Example:

```
String query = "INSERT INTO student(id, name, age) VALUES (?, ?, ?)";
PreparedStatement ps = con.prepareStatement(query);
ps.setInt(1, 1);
ps.setString(2, "Nikhil");
ps.setInt(3, 22);
ps.executeUpdate();
```

Read :-

- This operation is used to **fetch or read data** from a database table.
- In JDBC, we use a **SELECT SQL statement** to perform this action.
- The SELECT query is executed using a **Statement** or **PreparedStatement** object.
- The result of the query is stored in a **ResultSet** object.

- **ResultSet** helps to:
  - Move through each row of data one by one.
  - Access individual column values (like name, age, etc.).

- Example:

```java
String query = "SELECT * FROM student";
ResultSet rs = st.executeQuery(query);
while (rs.next()) {
    System.out.println(rs.getInt("id") + " " + rs.getString("name") + " " + rs.getInt("age"));
}
```

Update :-

- This operation is used to **change or modify existing records** in a database table.
- In JDBC, we use an **UPDATE SQL statement** to perform this action.
- Usually, a **WHERE clause** is added to specify **which record(s)** should be updated.
- Without a WHERE clause, all rows in the table could be updated by mistake.
- The UPDATE query is executed using a **Statement** or **PreparedStatement** object.

- Example:

```
String query = "UPDATE student SET age = ? WHERE id = ?";
PreparedStatement ps = con.prepareStatement(query);
ps.setInt(1, 23);
ps.setInt(2, 1);
ps.executeUpdate();
System.out.println("Record updated successfully!");
```

Delete :-

- This operation is used to **remove or delete records** from a database table.
- In JDBC, we use a **DELETE SQL statement** to perform this action.
- A **WHERE clause** is usually added to specify **which record(s)** to delete.
- Without a WHERE clause, **all rows** in the table will be deleted accidentally.
- The DELETE query is executed using a **Statement** or **PreparedStatement** object.

- Example:

```
String query = "DELETE FROM student WHERE id = ?";
PreparedStatement ps = con.prepareStatement(query);
ps.setInt(1, 1);
ps.executeUpdate();
System.out.println("Record deleted successfully!");
```